Day 29 – BIP Report with Parameter and LOV (Oracle Fusion Cloud)

Topic: LOV (List of Values) for the Parameter

Prepared by: Kawther Ali Almandhari

Concepts

- **Parameter Report:** applying a filter to your report. The parameter is added to the data set SQL as a bind variable (for example :IA_NAME), and the user enters its value when running the report.
- **LOV (List of Values):** a list of values defined for the parameter. When the LOV is attached to the parameter (Parameter Type = Menu), the user selects the value from a drop-down list instead of typing it.
- **Bind variable:** a name that starts with a colon (:) in the SQL. When you click OK, BIP detects it and offers to create a parameter with the same name.

Getting Started

Open BI Publisher

1. Open the URL: https://fa-euth-dev50-saasfademo1.ds-fa.oraclepdemos.com and sign in.
2. Go to **Tools > Reports and Analytics**.
3. Click **Browse Catalog**.
4. In the Folders pane, open **My Folders > KA** (my working folder).

Task 2 – Create Report with Parameter and LOV

This task has two parts. Part A adds an LOV to the parameter IA_NAME created in the Parameter Report task (DM_ABSENCEMGMT_BU). Part B builds a new data model with a parameter and an LOV from the beginning, then creates a new report.

Part A – Add LOV to the Existing Parameter (IA_NAME)

A1. Create the LOV

1. Open **DM_ABSENCEMGMT_BU** again (Catalog > My Folders > KA > Edit).
2. In the left pane, click **List of Values**. The list is empty: "[This area is currently empty.]".
3. Click **+** (Create new). A new row appears with the default name New_List_of_Value 1 and Data Source = demo.
4. Change the row to:
   - **Name:** LOV_IA_NAME
   - **Type:** SQL Query
   - **Data Source:** Oracle BI EE

A2. Write the LOV query

1. **Note:** do not paste the full data set query into the LOV. The LOV query must return only the one column used for the list, and it must not include the WHERE :IA_NAME condition. If you pasted it, delete it.

2. Click **Query Builder**.
3. From the catalog Payables Invoices – Transactions Real Time, open **- Invoice Accounting** and check only **Ledger Currency Name**.
4. Click **Save** to return to the data model.
5. Add **distinct** after select, so that each currency appears only once in the list. The final LOV query:

```
select distinct "- Invoice Accounting"."Ledger Currency Name" as "Ledger
Currency Name"
from "Payables Invoices - Transactions Real Time"."- Invoice Accounting"
"- Invoice Accounting"
```

6. Click the **Save** icon. The message "Saved" appears.

A3. Attach the LOV to the parameter

1. In the left pane, click **Parameters > IA_NAME**.
2. Change **Parameter Type** from Text to **Menu**.
3. In the lower panel (IA_NAME: Type: Menu), set:
   - **Display Label:** IA_NAME
   - **List of Values:** LOV_IA_NAME
   - **Number of Values to Display in List:** 100
   - **Options:** Multiple Selection checked, Can select all checked (NULL Value Passed)
4. Click the **Save** icon.
5. **Tip:** if you want to choose more than one value with Multiple Selection, change the condition in the data set from = to IN: where "- Invoice Accounting"."Ledger Currency Name" IN (:IA_NAME)

A4. Run the report with the LOV

1. Go to **Catalog > My Folders > KA** and open **REPORT_ABSENCEMGMT_BU**.
2. IA_NAME is now a drop-down list (LOV). Select **Euro** and click **Apply**.
3. Result: the report shows only the rows where Ledger Currency Name = Euro (Ledger Currency = EUR).

Part B – New Data Model with Parameter and LOV, then Report

Order: 1) Data Model, 2) Parameter, 3) LOV, 4) Report

B1. Create the Data Model and the Data Set

1. From the top menu, click **Create > Data Model**. A new Untitled data model opens.
2. On the Diagram tab, click **+ (New Data Set)** and choose **SQL Query**.
3. In the New Data Set – SQL Query window, enter:
   - **Name:** DM_Departments
   - **Data Source:** Oracle BI EE (change it from demo (Default))
   - **Type of SQL:** Standard SQL
4. Click **Query Builder**. In Catalog, choose **Absence Management - Availability Patterns Real Time**.
5. Click the **Department** table and check: Department Name, Department Address, Location Code, Location Name.
6. Open the **Conditions** tab to review the selected columns, then click **Save**.
7. Add the WHERE condition with the bind variable at the end. The final data set query:

```
select "Department"."Department Name" as "Department Name",
       "Department"."Department Address" as "Department Address",
```

```
       "Department"."Location Code" as "Location Code",
       "Department"."Location Name" as "Location Name"
from "Absence Management - Availability Patterns Real Time"."Department"
"Department"
where "Department"."Location Name" = :LOC_NAME
```

8. Click **OK**.

B2. Create the Parameter

1. The **Add Parameter** window appears. Check **LOC_NAME** and click **OK**. The parameter LOC_NAME is created automatically.
2. In the left pane, click **Parameters > loc_name**. It has Data Type = String and Parameter Type = Text.
3. Set **Default Value** = **London**.

B3. Create the LOV

1. In the left pane, click **List of Values**, then click **+**.
2. Enter:
   - **Name:** LOV_LOCATION
   - **Type:** SQL Query
   - **Data Source:** Oracle BI EE
3. Click **Query Builder**. Catalog: **Absence Management - Availability Patterns Real Time**.
4. Click the **Department** table, check only **Location Name**, and click **Save**. The generated LOV query:

```
select "Department"."Location Name" as "Location Name"
from "Absence Management - Availability Patterns Real Time"."Department"
"Department"
```

B4. Attach the LOV to the Parameter

1. Click **Parameters > loc_name**.
2. Change **Parameter Type** from Text to **Menu**. (The other options in the list are Date, POV and Search.)
3. In the lower panel (loc_name: Type: Menu), set:
   - **Display Label:** loc_name
   - **List of Values:** LOV_LOCATION
   - **Number of Values to Display in List:** 100
   - **Can select all:** checked, then choose All Values Passed

B5. Test the Data Model and save sample data

1. Click the **Data** tab. The LOC_NAME field appears.
2. Enter (or select) **London** and click **View**. Wait for "Processing".
3. The Tree View shows the data: DATA_DS > LOC_NAME (London) > G_1 with Department Name, Department Address, Location Code and Location Name (for example HCM UK, 1 South Place, London, London).
4. Click **Save As Sample Data**. The message "Saved as Sample Data" appears. Click **OK**.

B6. Save the Data Model

1. Click the **Save** icon (top right).
2. Save it in **My Folders > KA** with:
   - **Name:** DM_PAR_LOV

- **Description:** DM FOR PAR AND LOV FOR DEPARTMENT FROM ABSENCE MGMT

3. The "Saved" message appears. Go to **Catalog** and check that DM_PAR_LOV is in My Folders > KA.

B7. Create the Report

1. From the top menu, click **Create > Report**. The Create Report wizard opens on the Select Data step.
2. Choose **Use Data Model**.
3. Click the **Search (browse)** icon next to Data Model.
4. In Select Data Model, open **My Folders > KA**, select **DM_PAR_LOV** and click **OK**. The Data Model field shows /~A.A/KA/DM_PAR_LOV.xdm.
5. Under "How do you want to create your report?", choose **Guide Me** and click **Next**.
6. In Select Layout, choose the **Chart and Table** style and click **Next**.
7. In Create Chart, use **Location_Name** as the value and **Department_Name** as the label (bar chart). Click **Next**.
8. In Create Table, keep the columns Department_Name, Department_Address, Location_Code and Location_Name. Click **Next**.
9. In Save Report, save the report in **My Folders > KA** with the name **RPO_LOV_PAR**. Click **Finish**.

B8. Run the Report and use the LOV

1. Go to **Catalog > My Folders > KA** and open **RPO_LOV_PAR**.
2. The parameter **loc_name** appears as a drop-down list (LOV). Select **London** and click **Apply**.
3. Result: the report shows a bar chart of the departments and a table of the London departments (HCM UK, Receivables UK, Operations UK, Consulting South UK, Manufacturing UK, Marketing UK, Commercial Sales UK, Government Sales UK, Sales UK, Tele-Sales UK, Legal UK, Human Resources UK, Payables and Assets UK, Consulting UK), all at 1 South Place, London.

Summary

- **Parameter:** add where <column> = :PARAM_NAME to the data set SQL, then click OK, check the bind variable, Save, and Apply the value in the report.
- **LOV:** List of Values > + > Name, SQL Query, Oracle BI EE > Query Builder (one column, use distinct if values repeat) > Save.
- **Attach the LOV:** Parameters > select the parameter > Parameter Type = Menu > List of Values = the LOV name > Save.
- **Report:** Create > Report > Use Data Model > browse to the DM > Guide Me > layout (Chart and Table) > Save > Open and Apply.